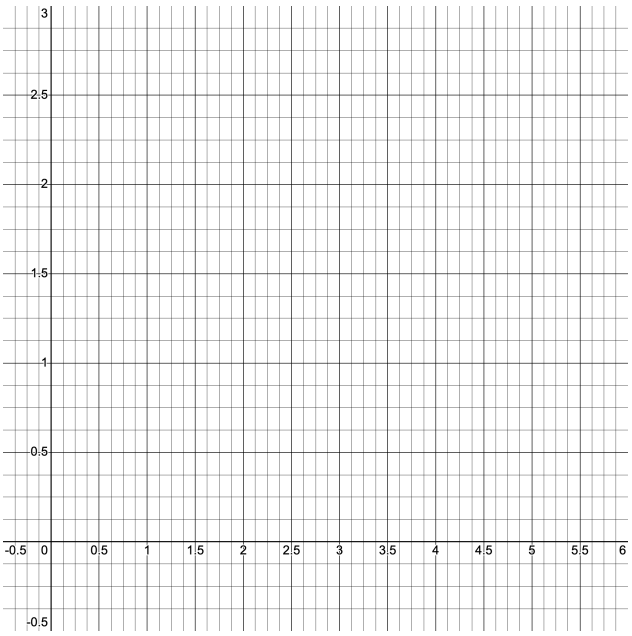
Phase 2 (part 1): Solving Linear Systems of Equations

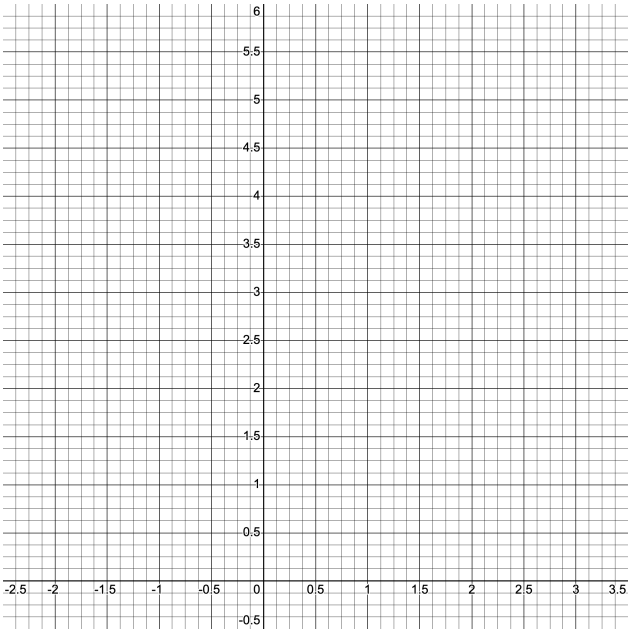
1.] Consider the following linear system:

$$x - 2y = 1$$
$$x + 2y = 5$$

a.) Solve the linear system for x and y using elimination. Then, graph each line below to confirm the solution is correct.



b.) Write the system as a linear combination, and confirm that the solution above is correct. Sketch the original vectors below and linear combination of the vectors on the plane below:



2.] Suppose we want to find the equation of the line $y = mx + b$ that goes through the points $(1, -3)$ and $(-4, 5)$. Find the equation of the line by first formulating the problem as a 2×2 linear system of equations. Then, solve it in Python and plot the solution along with the two points to show that the line goes through the points.

3.] Suppose we want to find the equation of the plane $z = m_1x + m_2y + b$ that goes through the points $(1, -3, 2)$ and $(-4, 5, 1)$, and $(0, 3, 1)$. Find the equation of the plane by first formulating the problem as a 3×3 linear system of equations. Then, solve it in Python and plot the solution along with the three points to show that the line goes through the points.